

EEGAVI: a Foundation Model for Physiological and EEG Signals for Emotion Recognition

Jacopo Fichera

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Abstract

Labeled information has shown itself to be effective in machine learning. Unfortunately the supervised learning paradigm comes with significant limitations as it is very time and resource consuming. There are also situations where labeling is not only hard but too unreliable to produce satisfying results. Adopting unsupervised learning techniques such as *contrastive learning*, we propose a new *foundation model* for electroencephalography designed to work in the domain of affective computing, *EEGAVI*. The approach is based on the assumption that different modalities encode complementary aspects of the same underlying phenomena. Beyond traditional EEG measurements, we therefore leverage additional modalities to enrich the learned representations. Datasets in the field are typically limited in size, so we enhance the model by integrating data from the VATE dataset through Knowledge Distillation with further information from video, audio and text modalities.

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1 Introduction

Machine Learning advancements can be applied across a wide range of domains due to their ability to model complex data. This has also influenced the field of affective computing where electroencephalography (EEG) is used to infer emotional and cognitive states. In recent years many models have been developed for EEG representation learning. These general-purpose models, *foundation models*, in the field tend to be limited by various reasons, with the low cardinality of EEG-centric datasets being one of them. This is especially noticeable once one considers the size of corpora of most text-based state-of-the-art (SOTA) models.

While multimodal foundation models have shown success in vision and language, their application to EEG representation learning remains limited. With this problem in mind, the goal of the project is to leverage cross-modal knowledge to integrate information into EEG representations. Our approach is limited to the affective aspect of neurological data processing, but the core idea can be applied to any field.

EEGAVI transfers information from multiple modalities into the EEG, aiming to enhance the representational capacity. It employs a principled integration of existing techniques (cross-attention + contrastive loss + knowledge distillation) applied to EEG under data scarcity. The model uses multiple frozen backbones to elaborate the input modalities. For the EEG data stream it is CBraMod[15].

Beyond the need for a foundation model, the research question focuses on the impact of *knowledge distillation* in the proposed context. In particular, we aim to investigate whether distillation in a multimodal context where sources and target do not belong to the same modality is of benefit to the training pipeline or not. To answer this question, we pretrain an instance of *EEGAVI* and analyze its performance using CBraMod as a reference point. These analyses are first performed on cross-retrieval metrics on the held-out data and later in various downstream tasks.

2 Related Work

2.1 EEG Foundation models

Foundation models are large-scale machine learning models trained on vast datasets. The ones that adopt self-supervised learning techniques typically learn general latent representations. These models are commonly used as backbones for downstream tasks. There have been various proposals for EEG foundation models. These, most times, suffer from limitations that are directly linked to the size and task specificity of their datasets. EEG measurements are very experiment-dependent and subject-specific. Unless a novel approach is able to abstract the subject-specific encoding inside the data, most solutions will encode the identity features in the learnt representations. For this reason, evaluations in the field are more commonly performed subject rather than experiment-wise. Beyond this, EEG experiments can vary much between different datasets. The measurement layout used for *AMIGOS*[10] with the *Emotiv EPOC Neuroheadset*, for example only tracks some of the electrodes tracked by *DEAP*[7] and its *BioSemi ActiveTwo EEG recording system*. It also runs at a different sampling rate. Thus, the cross-dataset compatibility tends to be lower than for other common tasks such as vision and audio.

CBraMod has been designed to tackle these problems as its representations are in fact not tied to the single electrode position but to the relative space the nodes occupy. This way the model learns geometry-aware representations of the data reducing the bias coming from the measurement setup. But it is not able to overcome the data limitations on its own. So, despite advances regarding spatial modeling, current EEG foundation models are still limited by dataset scale, inter-subject variability, and predominant unimodal training paradigms.

2.2 Multimodal Contrastive Models

Gathering more data is often a costly and difficult task. This motivates us to explore ways to exploit information from all available data modalities.

Such approaches have been widely adopted in many fields. *Flamingo*[2] by Google DeepMind is an application that partially inspired this work. It is a Visual-Language model that uses two frozen backbones for image and text generation. It leverages information given by the visual encoder to enrich the textual representations. The model requires the two data sources to be interleaved to encode the spatio-temporal information that derives from the visual encoder inside the corresponding segments of text. The visual tokens generated by the corresponding backbone are used to condition the text with a *cross-attention* mechanism[14]. This is applied in an interleaved way between the layers defined inside the frozen LM and the new

architecture.

This way, the model is able to preserve the capabilities of its *LM* while injecting the multimodal information progressively.

A commonly adopted alternative is CLIP[11] style models which resemble more our approach. Instead of fusing the information in the whole architecture the alignment is performed only at the embedding level.

It is safer regarding overfitting as data limitations or noise in *Flamingo* can limit the model’s generalization performance. CLIP also solves a problem posed by *Flamingo* in the domain of this application: it is completely unsupervised. CLIP style losses train the model on the batch level. Their objective is to align representations from two different modalities within a shared embedding space. Their effectiveness is therefore dependent on the batch composition.

The *Momentum Contrast (MoCo)*[6] model proposed a solution to solve limitations posed by CLIP style losses and contrastive learning. It builds a dynamic dictionary with a queue and a moving-averaged encoder. It reduces the memory load of the model during training by generating a queue of computed samples to run the contrastive loss on. This way, when in constrained environments, the *batch size* limits can be mitigated by having a representation proxy of the past processed samples. To avoid the representations from drifting between iterations the framework makes use of a moving average instance of the model.

While multimodal contrastive frameworks have demonstrated strong cross modal alignment, their applications on EEG representation learning remain limited, particularly under data scarce conditions.

2.3 Knowledge Distillation

Knowledge Distillation (KD) is a simple and widely adopted technique. Its idea is to leverage a teacher in a process where its learnt representations are mimicked by a target student. Its most common use case is for model compression, reducing the parameter count and computational cost.

Distillation as a framework can be applied at different points of a model’s architecture. It can also happen in a recursive fashion as self-distillation.

KD introduces an auxiliary signal that guides the optimization process during training, acting as a secondary loss term. This way the information contained in the teacher is transferred indirectly to the student. The student model is trained to approximate the teacher’s representations, thereby transferring higher-level structural information into a more constrained architecture. For the distillation to work, it is necessary that both models process aligned data, or else alignment would be pointless. VATE[1], a multimodal dataset for video, audio and text in the affective computing domain, provides a suitable teacher source for cross-modal distillation into EEG representations.

The technique in cross-modal applications is being explored more and more in many contexts. However, its application to EEG representation learning is still limited even when multimodal data are available. For this reason, we investigate cross-modal distillation in the EEG domain.

Taken together the integration of these techniques is not sufficiently explored yet.

Dataset	Subjects	Experiments	EEG	Video	Audio	Text	ECG	Samples
AMIGOS	40	16 + 4	✓	✓	✓	✓	✓	22k
EAV	42	200	✓	✓	✓	✓	✗	40k
DEAP	32	40	✓	✓	✗	✗	✗	12k
MAHNOB	29	20(1*)	✓	✓	✗	✗	✓	1k

Table 1: Datasets used for training summary. MAHNOB is missing most experiments as only few for participant are accessible.

3 Methodology

EEGAVI is a multimodal architecture built on frozen backbone encoders for each modality. The frozen representations are further elaborated before getting to a *cross-attention* mechanism. *EEG* is the main modality of the model acting as a *pivot* while the others are *supports*. This structure allows the model to generalize well even with limited or absent supporting modalities. Like many other representation models, *EEGAVI* outputs a *CLS* token that condenses sample information in a tighter space. It is used during training for contrastive alignment. The model’s default implementation is able to process EEG, video, text, audio and ECG data.

3.1 Data

For the development of the model, we use data from different multimodal datasets. Most of these lack some modalities that the model can process.

- **AMIGOS:** The dataset was composed for the research on affect, personality traits and mood by means of neuro-physiological signals. The experiments were performed on a total of 40 participants and are divided in two different subsets. For the first part, 16 short emotional videos from famous movies were shown to each participant alone. For the second one, subjects watched 4 long videos, some alone and some in groups. Self-assessments and external annotations on affective levels and emotions were also tracked.
- **EAV[8]:** *EEG-Audio-Video Dataset for Emotion Recognition in Conversational Contexts (EAV)* measures 30-channel EEG, audio and video from 42 participants. The scenarios proposed are done in a cue-based conversation designed around five distinct emotions. Each participant contributed in a total of 200 interactions. Participants tracked self-assessments for the emotion of the experiment (happy, calm, angry and sad) and their level of arousal. They were also evaluated by the experimenters on the same protocol.

Dataset	Train (70%)	Validation (10%)	Test (15%)
AMIGOS	30	4	6
EAV	32	4	6
DEAP	24	3	5
MAHNOB	22	3	4

Table 2: Partitioning of the subjects in the three data splits for training *EEGAVI*.

- **MAHNOB-HCI**[13]: is a broad dataset recorded in response to affective stimuli. Unlike other datasets *MAHNOB* also tracks gaze signals, but it is too limited in size to use. We did not include an explicit gaze stream due to a lack of calibrated gaze targets and potential estimation noise from frontal videos. However, eye- and head-related cues could be extracted as an auxiliary modality in future work.
- **DEAP**: *Database for Emotion Analysis* is a dataset composed of data from 32 participants. Each watched 40 one-minute long clips of music videos. Subjects rated their valence, arousal, like/dislike, dominance and familiarity levels. Audio for these is not given for copyright reasons.

All datasets are partitioned in train, validation and test set around (70-10-15)%. As commonly done for EEG applications, the partitioning is not performed on the sample level but on the participants directly, meaning the real split proportions might be a little off if data is corrupted, missing or a participant just made fewer experiments. This should allow the model to force a stronger cross subject generalization. Partitioning of the data is reported in table 2.

3.1.1 Sampling Strategy

Affective responses unfold across multiple temporal scales, for this reason the model must generalize well on both short events and longer temporal dynamics. While short clips may capture localized affective outbursts, long ones can encode the evolution of a subject’s perception over time. For the model to be robust across scales we sample variable-length EEG segments between 1 and 32 seconds using an EEG-centric sampling strategy.

Segment durations are sampled from a log-uniform distribution, favoring shorter clips. A sample is either taken at random from the clip with probability p or selected as a point of interest from the signal with probability $1 - p$. In the experimental setup $p = 0.6$. When an *anchor* point is selected short segments are built by left-aligning the window at the anchor time rather than centering it. This design may introduce a slight potential positional bias in short clips; however,

Class	Max length (s)	Same class IoU	Cross class IoU	Jitter Frac
Short	4	0	0.7	-
Medium	16	0.35	0.7	0.3
Large	32	0.25	0.7	0.4

Table 3: The jitter fraction indicates the maximum relative displacement applied to a segment’s center.

longer segments are centered on previously extracted short ones and therefore do not have this behavior.

If the resulting segment respects the overlap constraints defined for that class it is stored as a sample, else discarded. Segments are of three different settings as reported in table 3. The process is iterated until it fails to present new valid segments after running out of patience.

The candidate extractor first computes STFT-based frame features per EEG channel. The points of interest are extracted in the STFT domain following a set of heuristics commonly used when processing EEG. These anchors are identified through a multi-descriptor scoring mechanism that combines the following indicators:

- *Relative band-power per frame*: Fraction of total spectral power contained in predefined frequency bands (0.5–4, 4–8, 8–13, 13–30 Hz). Sudden dominance of a specific band indicates a change in oscillatory regime contributing positively to salience.
- *Spectral entropy*: Shannon entropy of the normalized power spectrum; high when the spectrum is flat (distributed energy), low when concentrated. Lower entropy is favored as it indicates structured spectral content rather than broadband noise.
- *Spectral flux*: Frame-to-frame positive spectral change computed on magnitude spectra, capturing rapid spectral dynamics. Rapid spectral changes are rewarded in the saliency extraction process.
- *Root-mean-square (RMS)*: Measures the average signal energy in a window, independent of frequency content. Higher energy contributes positively to candidate selection as it highlights strong neural responses.
- *Line length*: It is a simple but powerful time-domain EEG feature that measures the total amount of change in the signal within a window. It computes the average absolute first difference, acting as a proxy for signal complexity. In EEG it is often associated with high activity as seizures/epilepsy or motion artifacts, therefore its contribution is moderated in the scoring process.

- *Teager-Kaiser Energy Operator (TKEO)*: It captures bursty, high-frequency energy. Sensitive to rapid changes, it is popular in EEG analysis; adds robustness to the process as it might capture patterns not recognized by RMS alone.

Afterward, each channel’s features are standardized with a rolling median/MAD estimator with a 30s window. This is done to prevent high-variance features from dominating the saliency score, mitigating the effect of outliers. A saliency score per channel as a weighted sum of z-scored features is formed. These scores are aggregated by taking the mean of the top k channels. Artifact annotations are used to downweight candidate frames before anchor selection. Lastly, a greedy top-score selection with a minimum time gap constraint is applied to select the points.

3.1.2 Preprocessing

All modalities are processed individually.

For memory efficiency, the samples are passed through the frozen backbone of each modality directly in preprocessing. Given this limitation, it is impossible to compute training augmentations on demand, thus the idea was dropped entirely. Pre-processing is done for both *EEGAVI* and *VATE*, each with their own pipeline.

Given the intrinsic temporal structure our data has, a sample’s modalities are partitioned into smaller time blocks on which the embedding model is called. To avoid the excessive storage growth, sequence resampling is performed to generate segments that will act as temporal unit tokens.

All supported modalities are also symmetrically quantized in *int8* per dimension with a per-dimension scale vector to further reduce the space load.

Knowing that media can occasionally miss and that segments following our sampling strategy can vary in size, all modalities of a single sample are masked.

EEG (pivot) The main modality of the model.

As described when introduced, *CBraMod* works on the relative spatial arrangement between the measuring electrodes. So, for it to work correctly, each dataset must be first mapped into a common channel layout. We call this the canonical order, and it becomes part of the model definition as it has to be consistently applied across datasets. The canonical order, composed of 34 different EEG channels, used by the project is reported in table 4. All datasets are mapped into this canonical channel order; missing electrodes are zero-filled and masked.

The signal must also to be resampled ($fs = 200$) to generate one-second tokens by *CBraMod*. A sample can miss some channels, for this reason EEG data is masked with a boolean tensor and initialized to a zero vector.

Conceptually, *EEGAVI* is able to handle a variable number of input channels because it flattens the generated channel representation with the last dimension. For EEG handling and preprocessing, we use the *MNE* library.

For *EEGAVI* to work, all supporting modalities have to match the same time unit size. Their time unit is set to 4 seconds. This is not necessary for *pivot* whose time unit is 1 second.

Video All videos have a similar setup where the image is centered on the face of the subject. For this reason, no cropping to the input videos was performed.

The selected backbone for the Video embeddings is ViViT¹[3]. To preserve spatial-temporal richness we use the model’s *last_hidden_state* output. The raw output is high-dimensional and memory intensive, a single sample is of the shape (16, 14, 14, 768) which is then repeated for T time steps. To solve this problem 2D pooling is applied on the (14, 14) frame with *stride* = 2 and *kernel* = 3.

Audio All audio, when present, is turned into a mono signal and resampled to match the required $fs = 16kHz$ by WavLM²[5]. It is primarily a speech/paralinguistic model which might limit generalization to non-verbal vocalizations. EAV tracks speech of the subject but AMIGOS’s audio doesn’t have the participant as direct interlocutor. Future work could evaluate alternative audio encoders or separate speech vs. reaction audio streams.

Text No dataset provides a textual transcript of the experiments. For this reason, we transcribe speech using *Whisper*³ by OpenAI[12]. The process is only possible when an audio track is available and the duration of the media is shorter than 5 minutes. Whisper should be able to process long clips but the need for timestamps on the extracted words makes this unfeasible.

The Text processing pipeline uses MiniLM⁴[16] to generate embeddings of the text. To provide richer representations the segments are extended beyond their time window. They receive information from past blocks while also some from the one immediately after. This is done because embedding models, textual ones in particular, are known to make stronger representations when context is broader.

ECG These measurements are often versatile, as it is possible to reconstruct more of the signal even if the measurements are limited. As ECG foundation

¹ViViT: google/vivit-b-16x2-kinetics400

²WavLM: microsoft/wavlm-base

³Whisper: openai/whisper-medium

⁴sentence-transformers/all-MiniLM-L6-v2

Index	Channel	AMIGOS	EAV	DEAP	MAHNOB
00	AF3	✓	✗	✓	✓
01	F7	✓	✓	✓	✓
02	F3	✓	✓	✓	✓
03	FC5	✓	✓	✓	✓
04	T7	✓	✓	✓	✓
05	P7	✓	✓	✓	✓
06	O1	✓	✓	✓	✓
07	O2	✓	✓	✓	✓
08	P8	✓	✓	✓	✓
09	T8	✓	✓	✓	✓
10	FC6	✓	✓	✓	✓
11	F4	✓	✓	✓	✓
12	F8	✓	✓	✓	✓
13	AF4	✓	✗	✓	✓
14	Fp1	✗	✓	✓	✓
15	FC1	✗	✓	✓	✓
16	C3	✗	✓	✓	✓
17	CP5	✗	✓	✓	✓
18	CP1	✗	✓	✓	✓
19	P3	✗	✓	✓	✓
20	PO3	✗	✗	✓	✓
21	Oz	✗	✓	✓	✓
22	Pz	✗	✓	✓	✓
23	Fp2	✗	✓	✓	✓
24	Fz	✗	✓	✓	✓
25	FC2	✗	✓	✓	✓
26	Cz	✗	✓	✓	✓
27	C4	✗	✓	✓	✓
28	CP6	✗	✓	✓	✓
29	CP2	✗	✓	✓	✓
30	P4	✗	✓	✓	✓
31	PO4	✗	✗	✓	✓
32	PO9	✗	✓	✗	✗
33	PO10	✗	✓	✗	✗

Table 4: The canonical EEG channel order used by EEGAVI is based on the 10-20 system. The table reports the presence of each channel across datasets. AMIGOS employs the Emotiv EPOC headset therefore it has less tracked electrodes.

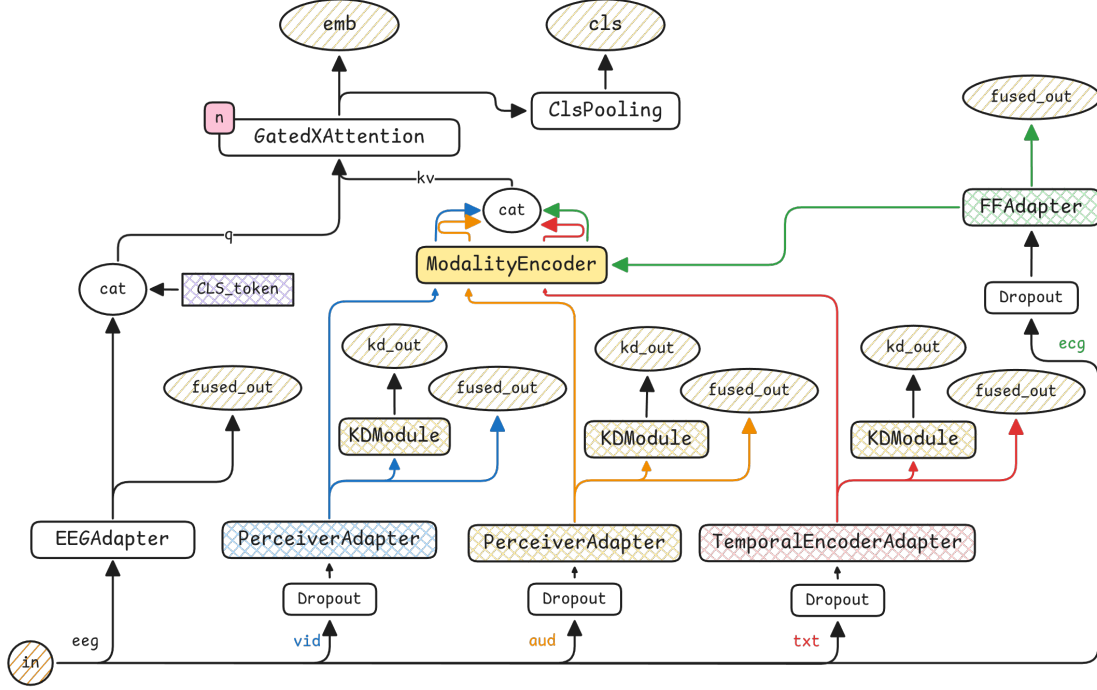


Figure 1: The model’s general architecture. Single custom blocks are reported at the end of the report in the resources section.

model, we opted for the ECG-FM[9] which is reported to perform when the data channels are incomplete.

3.2 Model

The model takes as input a sample that has a representation for each modality defined for the model. The core element of *EEGAVI* is the *gated cross-attention* module. The model performs pivot-centric cross-modal conditioning followed by intra-modal re-integration. Before passing through it, each modality follows a distinct flow that adds expressive power to the input embeddings.

This is to allow *KD* to affect the model during backpropagation, since the embeddings come from a frozen state. Having selected a contrastive model trained on *VATE*, the modalities that actively participate in the *KD* process are video, text and audio. Of these audio and video, given the way they are processed, are structurally complex. For this reason a more complex module to capture the information and condense them is used: the Perceiver Resampler. It was first introduced by Flamingo and its goal was to be able to elaborate all different kinds of shapes to standardize. In our case data is always of the same structure for the

way the inputs are masked, thus it only acts as data compressor. It uses attention mechanisms and a set of parameters to condense information.

The *KD* process happens before the concatenation of the supporting modalities and is applied to the pooled vector obtained from the "adapter" modules that come before the *ModalityEncoder* call. The student is drawn towards the teacher's representational space with SigLIP loss, defined later.

After the possible branching caused by *KD*, all modalities are tagged with a *modality encoder* and later stacked in a single embedding vector. The resulting vector acts as the key-value input in the *cross-attention* process while the EEG branch is the query. This way the conditioning is only directed by EEG data.

Since the model is masked on at all stages, to ensure stronger generalization capabilities, sample-modality level dropout is performed. Each sample of the batch with a set probability p can be masked out during the whole process downstream. This should allow the model to better manage data in situations where supporting modalities might be missing.

Structurally, the *cross-attention* module in EEGAVI does attention on the input query and key-value vectors and then proceeds to run *self-attention* to do global integration of the conditioned signals. This allows the information that was merged at different time steps to interact and propagate beyond local injections. Lastly, a feed-forward block is used to introduce a non-linearity, without it the process would behave as a structured linear mixer. Gating logic is also applied to every step of the module to control each step's contribution magnitude.

In the first definition of the model two *gated cross-attention* modules are stacked on each other.

3.3 Training Strategy

The model was trained using *AdamW* with a learning rate scheduler and an initial warmup process. To avoid the model becoming too biased towards the most numerous dataset in terms of samples we decided to build batches by following the power-law ($\alpha = 0.5$):

$$P(i) = \frac{n_i^\alpha}{\sum_j n_j^\alpha}$$

. This preserves the relative prevalence of larger datasets while also balancing their dominance through non-linear downweighting. For this reason, when referring to an epoch of the model the stochastic sampling is unlikely to make it an exact pass.

Samples tends to be very large because of the multiple modalities and their representation shape. For this reason training the model becomes memory demanding in large batch settings. While learning with small batches itself is fine and helps with regularization in contrastive setups it can be harmful. We adopt the *SigLIP*[18]

loss for training, a loss function that is inspired by InfoNCE.

$$l = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N \log \frac{1}{1 + e^{z_{ij}(-tx_i * y_j + b)}}$$

Generally speaking, SigLIP is less sensitive to batch size.

The reason for this is that SigLIP does not rely on softmax over in-batch negatives as it uses independent binary logistic losses. There is therefore no global softmax normalization, thus the gradient does not depend on the number of negatives.

While this helps, to make training even more robust MoCo was used. Standard values for the technique were initially used (momentum $m = 0.99$, queue size $q = 1024$) but will be fine-tuned in future work. The use of MoCo grants the model a larger number of negatives samples to compute the loss from, acting as a stabilizer for training.

Mixing data is generally a good practice but in the context of EEG it can be a limitation. We decided to make training batches built only from one dataset at a time. This technique allows training batches to present generally harder situations for the model to separate, a phenomenon shown to improve learning in contrastive setups. The negative samples drawn from the queue built by MoCo ignore the dataset. This allows the model to still combine samples coming from different datasets during the loss computation.

The inputs of the model define a temporal structure, this is an element controlled by the attention mechanism. Inspired by other applications such as AntLM[17], instead of taking single position of the time control we opt for a dynamic training on the setting. The model can elaborate the inputs in either *causal* or *bidirectional* setting. When *causal* is selected future time steps relative to the current in the attention mechanism are masked out, this way the model only has prior information to generate the outputs. The *bidirectional* setting allows for the whole sequence to be attended to. The AntLM model explicitly describes a unified framework that alternates between causal and masked in the field of language models. We take the same idea by balancing with a random probability p in what masking mode the next batch in training is. To make training simpler in earlier steps bidirectionality is preferred becoming more causal in later stages of the training process, controlled by a cosine ramp schedule.

		EEG_{fused}		EEG_{only}
		Val	Test	Test
Audio				
	MRR	0.96	0.98	0
	$Margin$	0.19	0.20	0
	$Top-1$	0.92	0.96	0
ECG				
	MRR	0.63	0.59	-7.05
	$Margin$	0.17	0.16	0
	$Top-1$	0.50	0.47	0
EEG				
	MRR	0.90	0.88	1.0
	$Margin$	0.31	0.31	0.51
	$Top-1$	0.85	0.84	1.0
Text				
	MRR	0.70	0.63	0.04
	$Margin$	0.14	0.15	0.00
	$Top-1$	0.47	0.50	0
Video				
	MRR	0.65	0.74	0
	$Margin$	0.27	0.29	-9.94
	$Top-1$	0.53	0.62	0
Mean				
	MRR	0.77	0.76	0.21
	$Margin$	0.21	0.0	0.0
	$Top-1$	0.66	0.68	0.20

Table 5:

4 First Results

The first training attempts of the model report the values shown in Table 5.

We applied early stopping on the global averaged value of mrr with patience of 12 epochs and a total limit of 40 epochs. The training was stable as reported in the loss graph running for a total of 40 epochs. The signal could still have improved over time.

The tracked metrics evaluate the results by measuring the capability to identify the originating modality from the fused vector. This idea is a robust method for understanding if the final embedding correctly condenses the information from the various modalities.

An often used metric to measure this behavior is *mean reciprocal rank* (MRR).

	AMIGOS	DEAP	MAHNOB	EAV
Audio				
<i>MRR</i>	0.97	-	-	1
<i>Margin</i>	0.16	-	-	0.21
<i>Top-1</i>	0.95	-	-	0.99
ECG				
<i>MRR</i>	0.60	-	0.42	-
<i>Margin</i>	0.16	-	0.15	-
<i>Top-1</i>	0.48	-	0.25	-
EEG				
<i>MRR</i>	0.92	0.98	0.97	0.99
<i>Margin</i>	0.26	0.35	0.30	0.32
<i>Top-1</i>	0.88	0.97	0.96	0.98
Text				
<i>MRR</i>	0.40	-	-	0.93
<i>Margin</i>	0.07	-	-	0.19
<i>Top-1</i>	0.17	-	-	0.87
Video				
<i>MRR</i>	0.55	0.76	0.78	0.81
<i>Margin</i>	0.17	0.33	0.23	0.29
<i>Top-1</i>	0.41	0.64	0.65	0.71
Mean				
<i>MRR</i>	0.69	0.88	0.72	0.93
<i>Margin</i>	0.21	0.34	0.27	0.33
<i>Top-1</i>	0.58	0.81	0.62	0.89

Table 6: Test results on each dataset independently.

It is a statistic that measures ranking systems by averaging the inverse rank of the first relevant result across multiple queries. The value is defined in the range $MRR \in [0, 1]$ with 1 indicating consistently being *top1*:

$$MRR = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \frac{1}{rank_i}$$

It is commonly used in information retrieval and scenarios where only the first correct answer matters.

Text exhibits significantly lower margin and MRR values, suggesting weaker cross-modal alignment. Margin measures the relative separation between positives and negatives pairs in the embedding space.

Beyond the *MRR* and *margin* we also calculate the *top-k* behaviors of the data. All measurements depend on the size of the sample. During training the value is moderated by the *MoCo* queue and the batch size while in validation and test it is the size of the dataset partition. All results of the measured metrics are reported in table 5. For reference all modalities are compared with an EEG-only version to see if fusion helps in the process of retrieval.

A hyperparameter tuning process may improve reported performance.

	Default	Term
α	1.0	Weight of the fusion loss term
β	1.0	Weight of the KD loss term
queue	1024	Size of the MoCo queue
xattn	2	Layers of <i>cross-attention</i>
lr	1.5×10^{-4}	Learning rate
b	64	Batch size
p	0.1	Dropout p for supporting modalities

Table 7: Hyperparameters to tune and their default value.

5 Next steps

5.1 Hyperparameter Tuning

Exhaustive tuning on low-level architectural parameters is avoided to favor a limited set of high-leverage hyperparameters. This is because the goal of the research is not to produce a *State-of-the-Art (SotA)* model but rather to verify the robustness of the proposed design choices. We ask ourselves if *KD* provides gains to the training pipeline and if multimodal alignments help to create richer representations of the data.

The hyperparameters to tune are reported in table 7. The α and β terms are used to balance the loss which is aggregated from the fusion and *KD* as:

$$\mathcal{L} = \alpha \mathcal{L}_{fusion} + \beta \mathcal{L}_{kd}$$

To reduce the search space we could fix α and only tune β .

5.2 Ablation

Once the hyperparameters tuning process is concluded, ablation is performed to assess the research question. The elements we focus on for the process are *KD* and modality fusion.

Knowledge Distillation We assume the technique helps with the learning process by integrating information encoded in the structure of the contrastive model of *VATE*. To see if the process really is of benefit we ablate it entirely. This allows us to quantify its contribution.

We also ablate each modality independently, since text in the initial training experiments showed poor performance on our evaluation metrics. If the teacher model has a weak performance on a modality, distillation from that branch may introduce noise rather than useful supervisory signal.

Modality fusion Not all modalities showed strong performance in the initial training experiments. It is possible that some modalities are too noisy or missing too often to extract information from. For this reason we remove one modality at a time to evaluate its individual contribution to the fused representation.

To assess whether fusion enriches EEG representations we compare the full model with a version trained using EEG representations only.

To further see if the contrastive alignment is strong we ablate the MoCo process as well. This ablation verifies whether improvements are due to improved negative sampling rather than multimodal integration alone. We expect a drop in performance considering its contribution during training.

5.2.1 Downstream tasks

To really measure the effectiveness of the modality fusion we have to apply the trained model to a downstream task.

For a first approach we will produce a simple predictor on *Valence-Arousal* using *EEGAVI* as frozen backbone. Given that we use all the data we can in the training of *EEGAVI* we have to exploit its validation and test sets. By having *EEGAVI* as frozen backbone, this does not lead to data leakage. We also integrate data from *DREAMER*, a dataset composed of EEG and ECG recordings that was not used in the training of *EEGAVI*.

We expect *EEGAVI* to outperform *CBraMod* on the task for two reasons:

- *EEGAVI* was pretrained on data drawn from the same sources
- The integration of multiple modalities favors richer embeddings than those computed by *CBraMod*

Once this experiment is performed, a more complex one will be done on "A Multimodal Virtual Reality Data Acquisition Platform and Dataset to Assess Systemic Human Cognitive States" [4]. The dataset is designed to measure human cognitive states in virtual reality environments. The dataset focuses mostly on cognitive workload, mental fatigue and stress, metrics it tracks from self-assessments of the subjects. We train two downstream predictors using *EEGAVI* and *CBraMod* as frozen backbone and compare their performances. Our analysis aims to see if *EEGAVI* was able to encode richer information on the affective cognitive state of a subject by the integration of multimodal information.

For future work a completely *Flamingo*-like approach could be studied. The current implementation is simpler as the fusion is only done at the end of the model's architecture. Interleaving layers of *CBraMod* with cross-attention blocks could result in a stronger model as we feed the information gradually during the generation of the embeddings.

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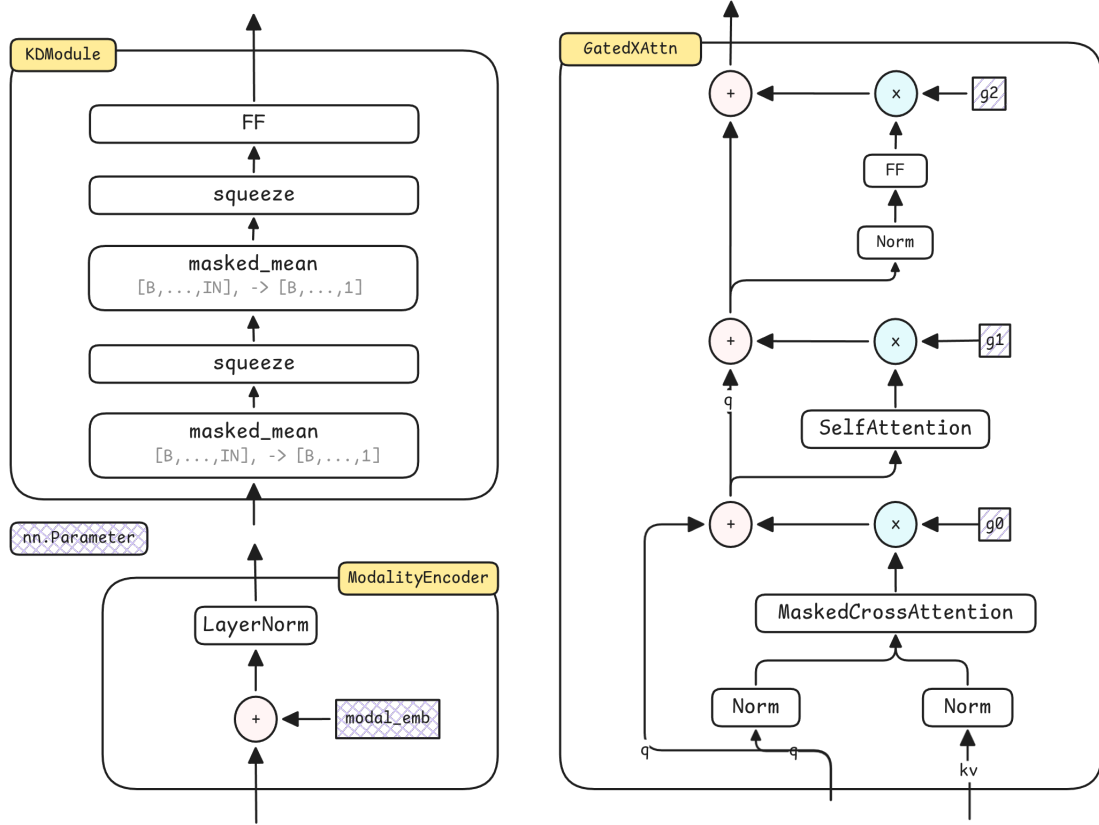


Figure 2: The *Knowledge Distillation* and the *gated cross-attention* blocks.

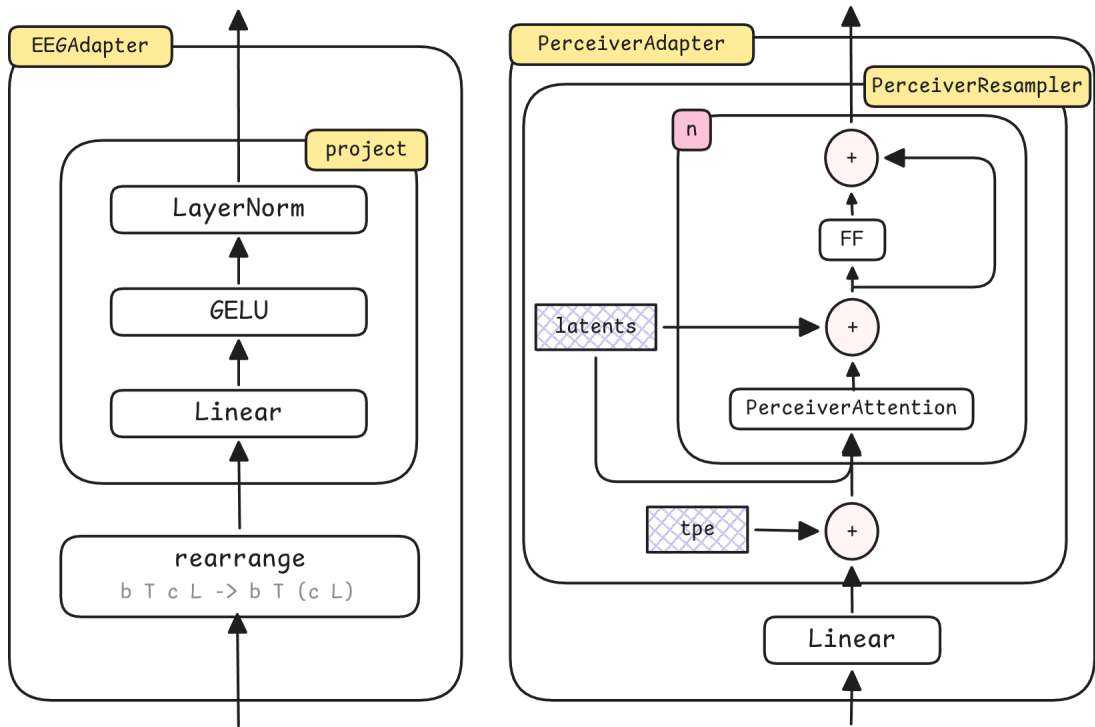


Figure 3: The EEGAdapter, used by the EEG branch, and the PerceiverAdapter used by video and audio streams.

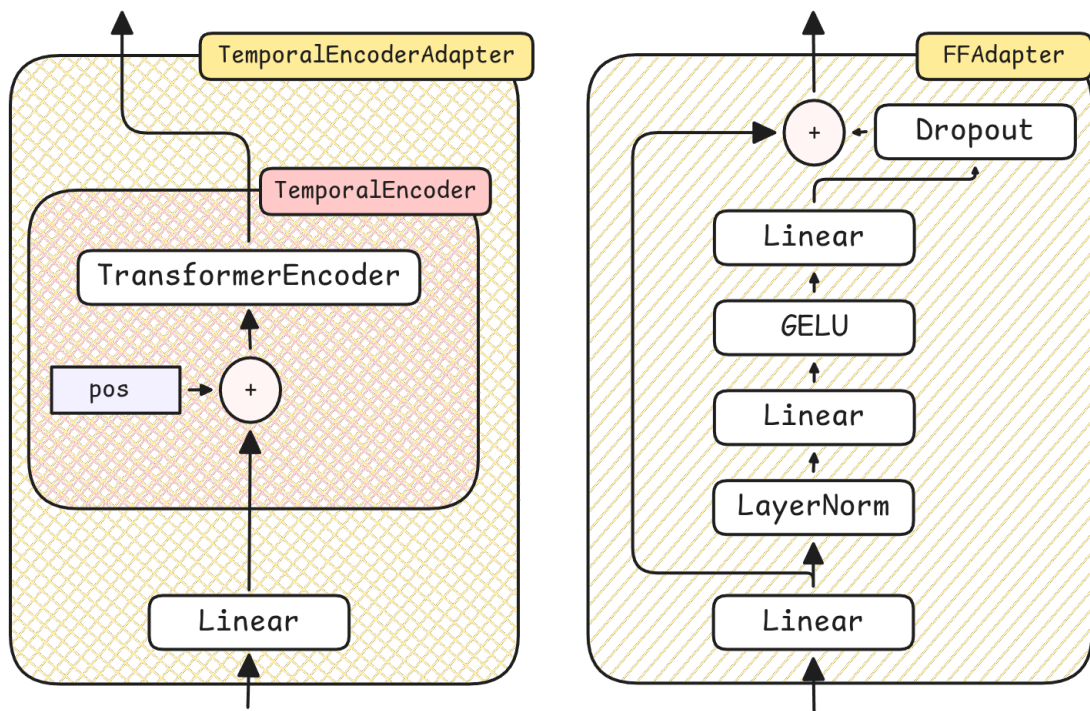


Figure 4: Adapters used by ECG and text branch as seen in the model’s architecture graph.